

Oluwaseun Omodele

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EDUCATION

University of Houston

Bachelor of Science in Electrical Engineering

3.5/4.0

Houston, TX

Expected May 2029

Organization: National Society of Black Engineers (NSBE), Member, Sept. 2025 – Present.

TECHNICAL SKILLS

Programming: Embedded C/C++, C#, Python, MATLAB

Embedded Systems: ESP32, Arduino, I2C, UART/USB serial, PWM, firmware debugging

CAD and Robotics: Fusion 360, SolidWorks, Unity, inverse kinematics, servo calibration

Prototyping and Test: FDM 3D printing, soldering, circuit troubleshooting, multimeters, Git/GitHub

PROJECTS

4-DOF Robotic Arm | *Fusion 360, ESP32, C/C++, Unity*

July 2026 – Present

- Engineered a custom **4-DOF robotic arm with a servo gripper**, integrating FDM-printed mechanisms, ESP32 firmware and PCA9685 PWM control across **5 servo channels** to execute pick-and-place tasks.
- Reduced prototype wobble and improved joint fit** by redesigning arm links, widening the base and refining printed clearances in Fusion 360; iterated the mechanical assembly for more stable motion.
- Developed a Unity interface around a **43-body CAD assembly**, integrating **inverse kinematics**, Xbox input and USB serial control to translate simulated joint motion into commands for the physical arm.
- Enabled **hands-free replay of recorded motion** through timestamped joint-command capture and playback; implemented a **500 ms communication watchdog** to hold the last commanded position when control updates stop.

002 DC-DC Buck Converter | *KiCad, ngspice*

October 2026

- Designed a **10 W buck converter** targeting **9–18 V input and 5 V / 2 A output**; selected a TPS54302-based topology and component values using datasheet analysis.
- Routed a **54 × 34 mm, two-layer PCB** with **16 components and 36 connected pins**; verified schematic-to-board connectivity and achieved **zero KiCad ERC/DRC violations** under configured rules.
- Analyzed startup, input variation and **0.5–2 A load steps** using an **educational ngspice switching model**; documented ripple and transient limits to guide further validation. **Not hardware tested.**

EXPERIENCE

Independent Online Reseller

Depop, eBay, Grailed, and Poshmark

Richmond, TX

May 2024 – Present

- Generated **\$2,000+ in revenue** across **4 marketplaces** by managing product photography, pricing, customer communication and order fulfillment.
- Applied **market research and condition-based pricing** to create accurate product listings; managed inventory, packaging and shipment preparation from listing through order fulfillment.

CERTIFICATIONS AND LANGUAGES

Certifications: CodeHS Python Level 1; Microsoft Office Specialist: Word

Languages: English, Yoruba